# 1. Start Here: Ratios, Cents, and Why Strings Matter

The two ideas you need for everything else: pitch ratios and the cent

## Pitch is a ratio, not a number

Two notes sound 'the same distance apart' whenever the ratio of their frequencies is the same. Going from 200 Hz to 300 Hz feels like going from 400 Hz to 600 Hz: both are a ratio of 3:2. Pitch perception is logarithmic, just like decibels.

## The cent: a logarithmic ruler

cents = 1200 × log2(f2 / f1). An octave (2:1) is exactly 1200 cents, so one equal-tempered piano semitone is 100 cents. Trained ears hear about 5 cents; our measurements across 13 masters agree to a few cents, so the cent is the right unit for comparing theory with performance.

## Why do simple ratios sound good?

If two tones have a ratio like 3:2, their harmonics line up (every 3rd harmonic of the lower note hits every 2nd of the upper one), so there is no 'beating'. The simpler the ratio, the smoother the sound. Ancient theorists discovered this with strings, long before anyone could measure frequency.

## Strings turn ratios into lengths (this is the key for a setar player)

On a string at fixed tension, frequency is inversely proportional to the vibrating length. Press a fret at 3/4 of the string and the pitch rises by 4/3; at 2/3 it rises by 3/2; at 1/2 you get the octave. A fretted instrument is therefore a physical ratio calculator: every fret position is a theory statement.

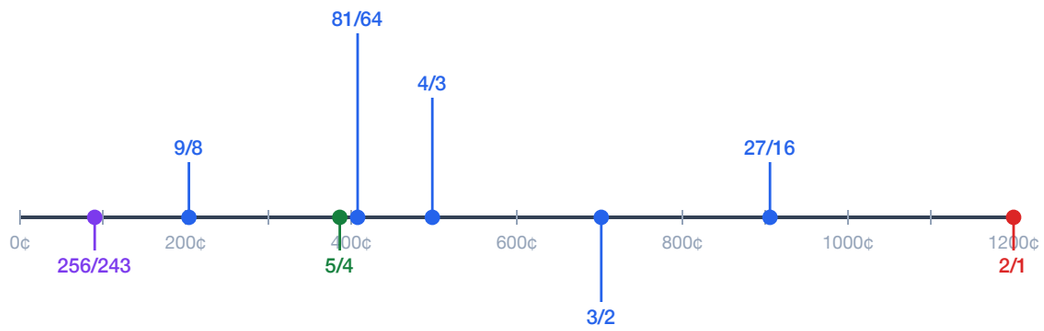
> Rule of thumb: ratio r above the open string → fret at 1/r of the length from the bridge, i.e. a distance (1 – 1/r) of the string from the nut. And stacking intervals means multiplying ratios (adding cents).

## How this document is organised


- Page 2 Pythagoras: the 3-limit system that everything else builds on or reacts to.
- Page 3 The setar as a multiplication machine (frets × strings).
- Pages 4–7 Al-Farabi, Ibn Sina, Al-Urmawi, Al-Maraghi: one page each.
- Page 8 Modern 20th-century theories (Vaziri, Barkechli, Farhat).
- Page 9 Side-by-side comparison with our Mahur measurements.
- Page 10 Dariush Tala'i's Theory of Dāngs & Setār Neck Architecture.
- Pages 11–15 Empirical modal consensus: Dashti, Navā, Bayāt-e Esfahān, Dastgāh-e Homāyūn, and Dastgāh-e Shūr.
- Page 16 Harmonizing microtones (Hannaneh quartal, Alizadeh drone, 13-limit spectralism).
- Page 17 Historical timeline, technical glossary, and further reading.


### A cents ruler for the common intervals

0–1200¢; dots = pure-ratio intervals, ticks = 12-TET semitones every 100¢




Blue = Pythagorean (3-limit) · Green = 5/4 just third · Purple = limma

Note how 81/64 (408¢) and 5/4 (386¢) differ by 21.5¢ — a distinction that will matter later.

### Ratio → cents → what 12-TET gives

The gaps between theory and the piano are the whole story

| Interval | Ratio | Just cents | 12-TET | Diff |
|---|---|---|---|---|
| Octave | 2/1 | 1200.0 | 1200 | +0.0 |
| Fifth | 3/2 | 702.0 | 700 | +2.0 |
| Fourth | 4/3 | 498.0 | 500 | −2.0 |
| Major 3rd (Pyth.) | 81/64 | 407.8 | 400 | +7.8 |
| Major 3rd (just) | 5/4 | 386.3 | 400 | −13.7 |
| Minor 3rd (Pyth.) | 32/27 | 294.1 | 300 | −5.9 |
| Whole tone | 9/8 | 203.9 | 200 | +3.9 |
| Limma | 256/243 | 90.2 | 100 | −9.8 |

# 2. Pythagoras (c. 570–495 BCE) and the 3-Limit System

Everything from pure fifths: octave, fifth, fourth, whole tone, limma, comma

## The idea

Tradition credits Pythagoras (Samos/Croton) with discovering that the consonances are small whole-number ratios on a monochord (a single stretched string with a movable bridge): octave 2:1, fifth 3:2, fourth 4:3. These use only the numbers 1–4 (the 'tetractys', 1+2+3+4 = 10), which the Pythagorean school considered sacred. The famous blacksmith-hammers story is a legend (it does not work physically), but the monochord facts are real. Written mathematics comes later (Philolaus, Plato's Timaeus, Euclid's Sectio Canonis).

## Building a scale from just 2 and 3

- Whole tone = fifth − fourth = (3/2)/(4/3) = 9/8 = 203.9¢.
- Two stacked tones: 9/8 × 9/8 = 81/64 = 407.8¢ (the 'ditone', a bright major third).
- What is left in the fourth: (4/3)/(81/64) = 256/243 = 90.2¢, the limma ('leftover').
- So a Pythagorean diatonic tetrachord is tone–tone–limma: 204 + 204 + 90 = 498¢ (a pure fourth).

## The catch: the Pythagorean comma

Stack twelve pure fifths and you land 23.46¢ sharp of seven octaves: 3^12 / 2^19 = 531441/524288 ≈ 1.0136. Pure fifths never close into a circle. That tiny leftover (the comma) is why every tuning system must make a compromise, and why Urmawi's 17-note idea (page 6) is interesting: he lets the chain run on instead of closing it.

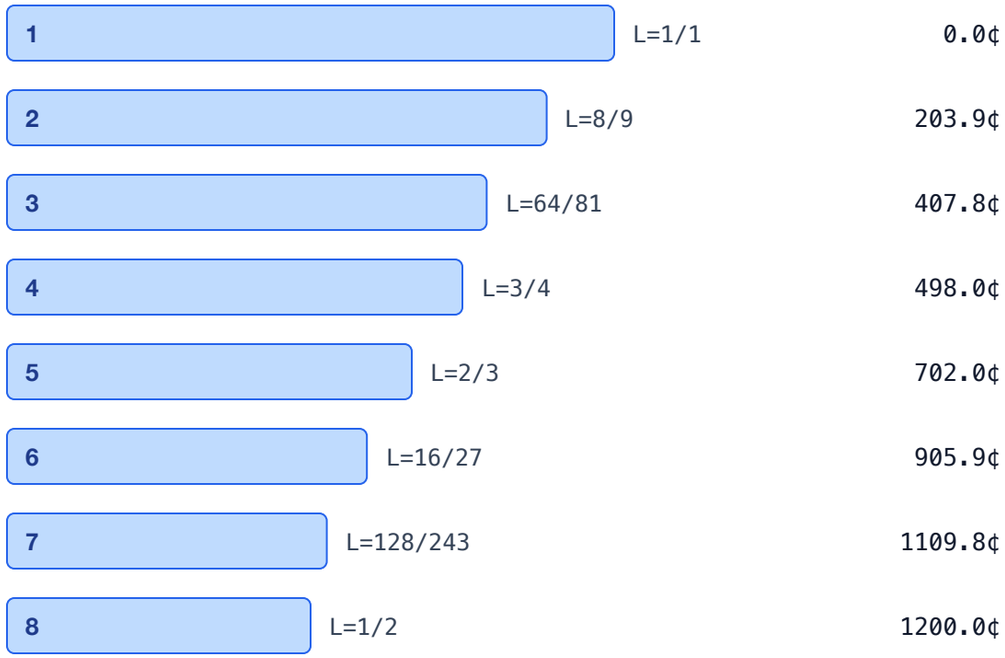
## Why an engineer should care

Pythagorean tuning is the unique 'closed under multiplication by 3/2 and 2' set: every pitch is 2^a × 3^b. That makes it easy to compute, easy to tune by ear (just a chain of pure fifths), and perfect for fretted lutes where one fret serves several strings. Its weakness: thirds (81/64) are 21.5¢ wider than the 'sweet' 5/4 of just intonation.

> Everything later in this document is either an extension of this 3-limit system (Urmawi, Maraghi) or a reaction to it (Zalzal's neutral third, Vaziri's equal quarter-tones).

### The monochord: one string, seven notes

Bar width = vibrating length (1 = full string). Shorter string → higher pitch.




Steps between bars: 9/8 · 9/8 · 256/243 · 9/8 · 9/8 · 9/8 · 256/243
(the Western major scale in Pythagorean tuning)

### Chain of fifths: how the comma appears

Each step ×3/2, folded into one octave (cents above the start)




Fifth #12 (red) lands 23.5¢ above the starting note: the Pythagorean comma.
Numbers = how many fifths were stacked.

# 3. The Setar as a Multiplication Machine

How one fret serves two strings, and why this matters for the whole analysis

## Pitch = open string × fret ratio

On a setar the open strings are tuned a fourth/fifth apart (we call the lower-register reference 'Do' and the string a fifth below it 'Sol↓'). Press a fret that raises the pitch by ratio r and the note is simply (open-string pitch) × r. In cents, you just add: cents(open) + cents(r).

## The shared-fret trick

A fret is a single physical object tied across the whole neck. The same fret pressed on the Sol↓ string (pitch 3/4 of Do) gives a different note than on the Do string. So with a handful of frets a player gets two layers of pitches, and the second layer is not free: it is locked to the first by multiplication.

Example from our Mahur study. The 'Re' fret (9/8) pressed on the Sol↓ string (3/4) gives 9/8 × 3/4 = 27/32 = –294¢ (the note we label La↓). A 'free' just-intonation sixth would be 5/3 → –316¢; the instrument forces –294¢, and the masters' measured La↓ ≈ –300¢ confirms the frets, not an ideal ratio, decide the pitch.

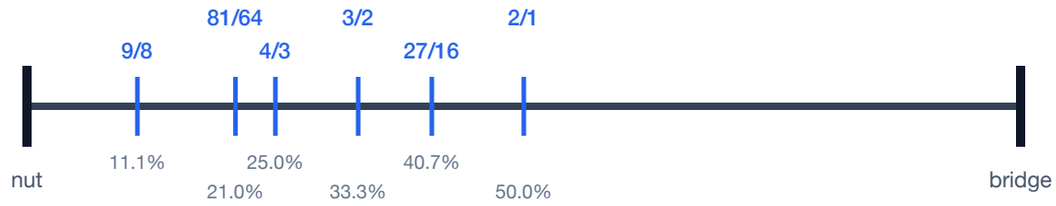
This is why measuring one fret on two strings is such a powerful test: the Pythagorean hybrid model predicts both numbers from one fret, with zero extra parameters.

## Practical tuning facts (engineer cheat sheet)

- Fret distance from nut = L × (1 – 1/r). For r = 9/8 that is 11.1% of the string; for 4/3, 25%; for 3/2, 33%.
- A 1¢ error = 0.058% of frequency; on a ~600 mm string, 1¢ at the 4/3 fret is about 0.1 mm.
- Moving the 'Re' fret by 10¢ moves every note derived from it on the other string by the same 10¢.

### Fret positions as fractions of string length

Distance from the nut (left) to the bridge (right); position = 1 – 1/r




### One fret, two strings (cents relative to Do)

Do string = open pitch 0¢. Sol↓ string = open pitch –498¢ (ratio 3/4).

| Fret ratio | On Do string | On Sol↓ string | Sol↓ ratio |
|---|---|---|---|
| 1/1 (open) | +0.0¢ | –498.0¢ | 3/4 |
| 256/243 | +90.2¢ | –407.8¢ | 64/81 |
| 9/8 | +203.9¢ | –294.1¢ | 27/32 |
| 32/27 | +294.1¢ | –203.9¢ | 8/9 |
| 81/64 | +407.8¢ | –90.2¢ | 243/256 |
| 4/3 | +498.0¢ | +0.0¢ | 1/1 |
| 3/2 | +702.0¢ | +203.9¢ | 9/8 |
| 27/16 | +905.9¢ | +407.8¢ | 81/64 |

Green numbers are the second layer produced for free by the same frets.

E.g. fret 9/8 on Sol↓ = 27/32 = –294¢ (our La↓); fret 32/27 on Sol↓ = 8/9 = –204¢ (Si↓).

# 4. Abu Nasr al-Farabi (c. 872–950): Frets, Zalzal, and the Neutral Third

Baghdad / Damascus · Kitab al-Musiqa al-Kabir ('The Great Book of Music')

## Who and why he matters

Al-Farabi was a polymath (philosopher, logician, musician) born in Transoxiana and active in Baghdad, Aleppo and Damascus. His Kitab al-Musiqa al-Kabir is the first systematic study of the 'ud and tunbur (long-necked lute, an ancestor of the setar) fretboard. He did what an engineer would do: catalogue every fret position used in practice and calculate their ratios.

## Key contributions

- Fret catalogue of the 'ud and the Baghdad and Khorasan tunburs, comparing the Pythagorean frets with those used by players.
- Documented the 'Zalzal' middle finger fret named after the 8th-century lutenist Mansur Zalzal: a neutral third near 27/22 ≈ 355¢, roughly halfway between minor (294¢) and major (408¢).
- Alongside it he lists the 'wusta of the Persians' (32/27 ≈ 294¢) and the Pythagorean 'binsir' (81/64 ≈ 408¢), so one hand position has three or four candidate middle-finger notes.
- Classified consonances (octave, fifth, fourth and their combinations) and the genera of the tetrachord, bringing Greek theory (Aristoxenus, Ptolemy, Euclid) into Arabic/Persian practice.
- Used 'super-particular' (epimoric) ratios (n+1)/n: whole tone 9/8 (204¢), minor tone 10/9 (182¢), semitone 16/15 (112¢), and microtonal neutral steps like 12/11 (151¢) and 13/12 (138.6¢) to express fret divisions.

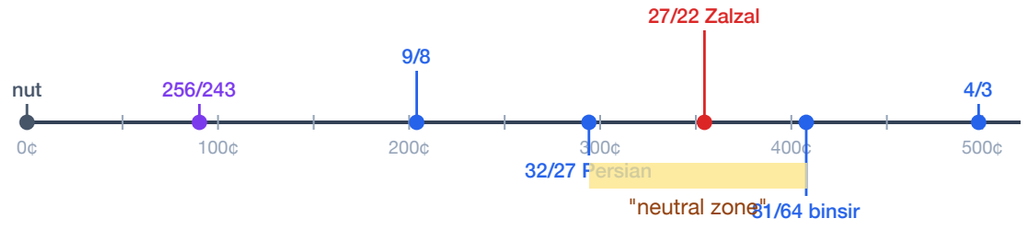
## Why this matters for our data: where does 13/12 come from?

To a modern ear trained on 5-limit ratios (2, 3, 5), 13/12 looks bizarre. But on an instrument string, dividing the distance between fret frets into equal geometric parts produces consecutive superparticular steps: 11/10 (165¢) → 12/11 (151¢) → 13/12 (138.6¢) → 14/13 (128¢). Medieval theorists (Fārābī and later Ibn Sīnā) analyzed these as the natural mathematical home for neutral steps. When we measure our master recordings, $Re^{k}$ sits right at +137.5¢ in Māhur and +138.7¢ in Chahārgāh — within 1¢ of 13/12, and 12¢ flat of Vaziri's artificial 24-TET 150¢!

Key takeaway: 13/12 is not an arbitrary modern curve-fit; it is the classical epimoric definition of a narrow neutral second (2/3-tone) documented in medieval lute fret analyses.

### Farabi's middle-finger candidates on the 'ud

Cents above the open string; the neutral zone sits between the two Pythagorean thirds




Red = Zalzal neutral third (≈355¢) · Blue = 3-limit frets · Purple = limma
The Zalzal note is 60¢ above the Persian wusta and 53¢ below the binsir.

### Timeline: who played what in the 'middle finger' slot

A thousand years of one finger

- **8th c.** Zalzal introduces the neutral middle fret on the 'ud
- **10th c.** Farabi catalogues Zalzal, Persian and Pythagorean frets
- **13th c.** Urmawi drops the neutral and closes the system on 3-limit (17 notes)
- **15th c.** Maraghi keeps Pythagorean frame; still describes older neutral usage
- **20th c.** Vaziri: equal quarter-tones; Barkechli/Farhat: measured neutral intervals

# 5. Ibn Sina / Avicenna (980–1037): Synthesis and the Persian-Language Treatise

Bukhara · Hamadan · Isfahan — music chapters of al-Shifa, al-Najat and Danishnama-yi 'Ala'i

## Who and why he matters

Ibn Sina is best known as a physician and philosopher, but he wrote three major music chapters: in al-Shifa ('The Healing'), in al-Najat ('The Deliverance') and, in Persian, in Danishnama-yi 'Ala'i ('Book of Knowledge for 'Ala' al-Dawla'). The Persian version is notable: it moves music theory from Arabic into the language of the Persian court.

## Key contributions

- Synthesis: reconciles the Greek mathematical tradition (Euclid, Aristoxenus, Ptolemy) with the practice-based Arab lute tradition of Farabi.
- Consonance theory: explains why intervals are consonant by ratio, ranks them and describes how larger ones are built from smaller ones.
- Genera of the tetrachord (jins): the diatonic, chromatic and enharmonic divisions of a fourth, including the soft 'equable' divisions that give neutral steps.
- Rhythm (iqa'): a formal treatment of rhythmic cycles that later became central in Urmawi's Adwar.
- Pitch vs. instrument: separates the abstract interval system from any one instrument's fretting.

## A tetrachord is a 4:3 box split into three steps

The tetrachord (jins) is the building block: a perfect fourth (498¢) divided into three steps. Where you place the two inner notes defines the 'genus' (flavour). The diagram on the right compares four divisions: the Pythagorean diatonic, Ptolemy's intense diatonic, Ptolemy's equable diatonic (neutral steps) and a 24-TET division.

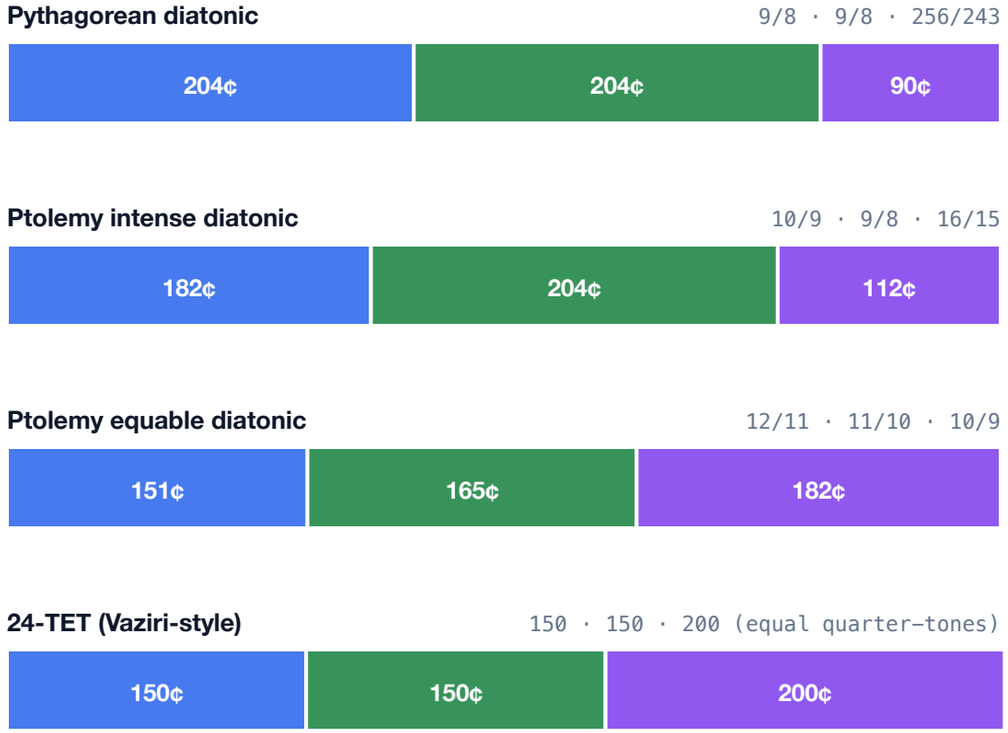
## Why this matters for us

In every dastgah each jins is a miniature 4:3 box. Our measurements show the outer boundary of each box (Do–Fa, Do–Sol) is very precisely 4:3 / 3:2, but the inner notes are where the schools differ.

Ibn Sina's importance for this project is conceptual: he fixes 'tetrachord = fixed outer fourth + flexible inner notes', which is exactly the structure we see empirically.

### Four ways to split a perfect fourth (498¢)

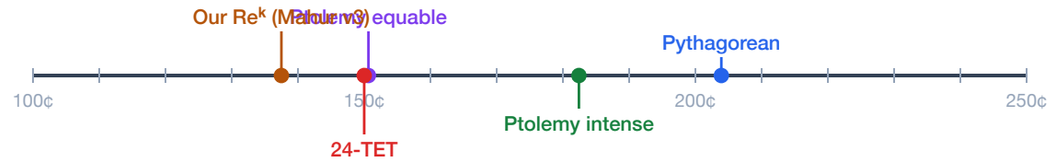
Bar segments = the three steps of the tetrachord, in cents




All four add up to 498¢; only the two inner notes move.

### What a neutral step looks like

Second note of each tetrachord, cents above the first




First step of the tetrachord: our measured Re$^k$ (≈137.5¢) sits with the small neutral steps, not with the 204¢ Pythagorean tone and not exactly on the 24-TET 150¢.

# 6. Safi al-Din al-Urmawi (c. 1216–1294): The 17-Note Pythagorean Octave

Urmia → Baghdad · Kitab al-Adwar (c. 1250s) and al-Risala al-Sharafiyya (c. 1267) — the 'Systematist school'

## Who and why he matters

Urmawi was a scribe, singer, lutenist and theorist at the court of the last Abbasid caliph in Baghdad; he survived the Mongol conquest (1258) and served the Ilkhanid vizier Juvayni, to whom the Sharafiyya is dedicated. His books are the foundation of what historians call the 'Systematist school' of Persian/Arabic music theory.

## Key contributions

- 17 notes per octave from a chain of pure fifths. Instead of closing the circle at 12, he lets the chain run to 17, giving two kinds of small step: the limma (90.2¢) and the Pythagorean comma (23.5¢).
- Fretting the 'ud so that all 17 notes of the octave can be found, replacing the ambiguous neutral frets of the earlier tradition.
- The mode system: 12 maqamat plus 6 avazat — the ancestor of our dastgah/avaz organisation.
- A rhythmic theory of cycles (adwar) and the first precise written notation using letters for pitch and rhythm.
- Your uploaded diagram: the double-octave drawn as 4:3 × 4:3 × 9:8 arches (2:1 = 4:3 × 4:3 × 9:8) from his manuscript tradition. Pure 4:3 fourths and a 9/8 tone are the whole skeleton.

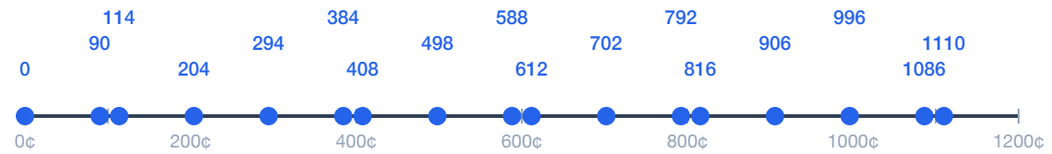
## What this means in our data

The outer skeleton we measure in Mahur (Sol↑ +702.1¢, Fa↑ +495.9¢, Re +201.5¢) matches the 3-limit frame within 0.1–2.4¢. The only deviation from Urmawi's frame is the major third: Nariman and Saba play 5/4 (+386¢), the 'just' third that appears as an alternative table in the Sharafiyya manuscript.

> Urmawi's lasting trick: treat the 17 notes as a menu. A given mode only uses 7–8 of them, and different modes pick different subsets of the same 17 positions.

### The 17 notes of the octave (folded chain of fifths, k = –8…+8)

Cents above the tonic; tick spacing shows limma vs comma steps




**Step sizes between neighbours: 23.5¢, 90.2¢**

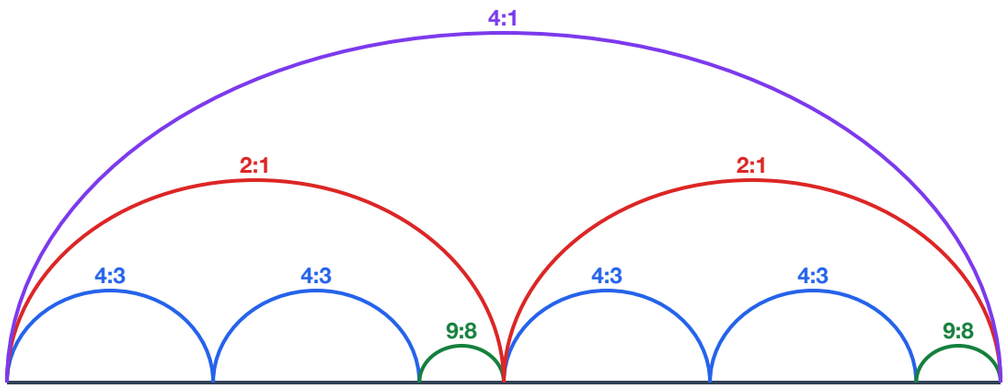
Limma = 90.2¢ (256/243) · Comma = 23.5¢ · Apotome = 113.7¢ (limma + comma)

Whole tone 203.9¢ = limma + limma + comma (90.2 + 90.2 + 23.5)

Chain extended to 17 fifths yields only two distinct step sizes: limma and comma.

### The double-octave skeleton from the manuscript

2:1 = 4:3 × 4:3 × 9:8, drawn as arches




0¢ … 498¢ … 996¢ … 1200¢ … 1698¢ … 2196¢ … 2400¢ (two octaves)

# 7. Abd al-Qadir al-Maraghi (c. 1353–1435): Persian-Language Systematist

Maragha → Tabriz → Baghdad → Samarkand → Herat · Jami' al-Alhan (1405), Maqasid al-Alhan (c. 1418)

## Who and why he matters

Maraghi was court musician under the Jalayirids and then the Timurids (Shahrukh in Herat). He is the great heir of Urmawi, but wrote in Persian rather than Arabic, which made his work the standard for later Persian musicians.

## Key contributions

- Kept and refined Urmawi's 3-limit frame: pure fifths, fourths and 9/8 tones, building every step from them.
- His preferred minor third is the Pythagorean 32/27 = 294.1¢ and major third 81/64 = 407.8¢ — both are 'frets in the 3-limit chain', not neutral pitches.
- Practical, performance-oriented theory: modes (maqam), shu'ab (offshoots) and complete compositions (naqsh, qawl, tarana, ghazal), so theory is tied to a living repertoire.
- Described the lute (ud) and its 'Persian' tunings so that fretted instruments could realise the whole system.
- Bridge to the Safavid period: his Persian terminology survives in modern dastgah vocabulary.

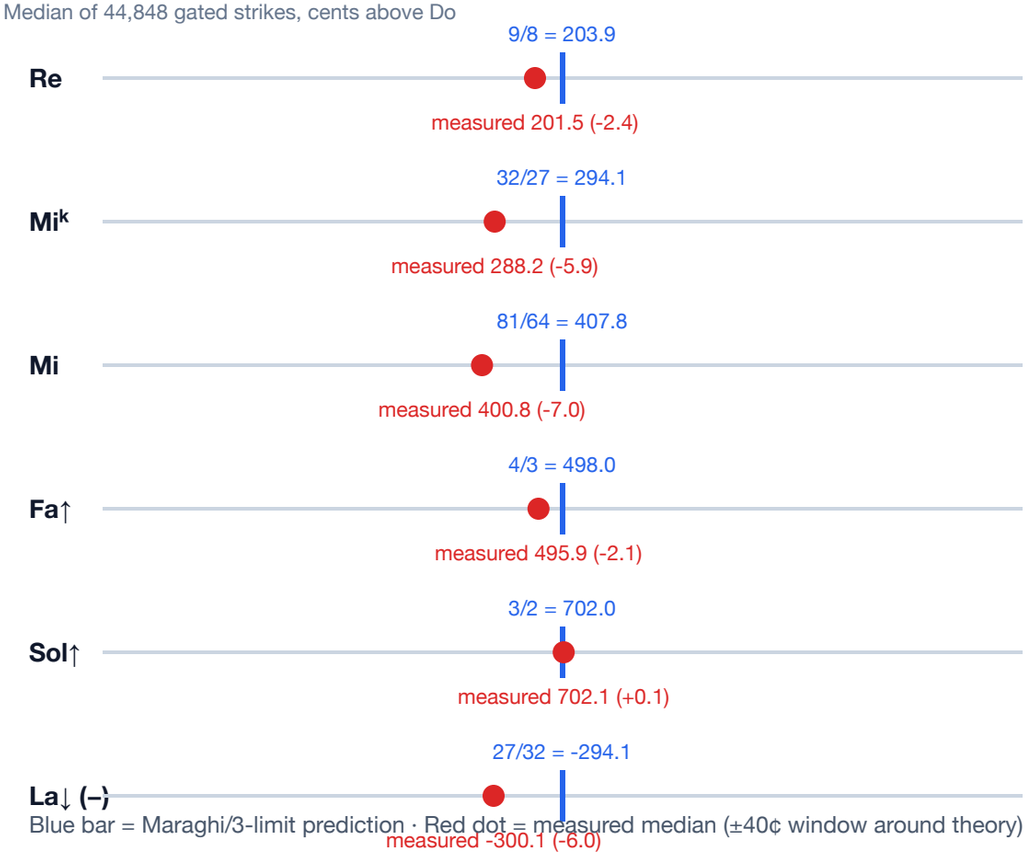
## Why this matters for our data

Our modern masters' Mi↓ (Mi$^{k}$ in Shekaste) ≈ +288.2¢ is just 5.9¢ below Maraghi's 32/27 (294.1¢) and nowhere near the old Zalzalian ≈340¢; the major third Mi ≈ +400.8¢ sits between his 81/64 (407.8¢) and Urmawi's 5/4 (386.3¢). The Qajar-trained masters (Borumand, Forutan, Neydavud) are the exception: their Mi$^{k}$ stays near 332–341¢.

Net result: modern Persian performance is closer to Maraghi's 3-limit frame than to Vaziri's equal-tempered quarter-tones, with two clear departures: the major third (scatter) and the old neutral third (survival).

### Maraghi's frame vs. what the masters actually play (Mahur v3)

Median of 44,848 gated strikes, cents above Do




Blue bar = Maraghi/3-limit prediction · Red dot = measured median (±40¢ window around theory)

### Same-fret check: Mi↓ on Do and Si↓ on Sol↓

One fret ⇒ two predictions

**Fret 32/27 on Do string → Mi↓ (Mi$^{k}$)**

theory +294.1¢ measured +288.2¢ Δ −5.9¢

**Same fret on Sol↓ string → Si↓**

theory −203.9¢ measured −211.0¢ Δ −7.1¢

Both come within ~6¢ of the Pythagorean prediction — the modern single fret is shared.

## Ali-Naqi Vaziri (1887–1981): 24 equal quarter-tones

Vaziri, who studied in Berlin, published Dastur-e Tar (1913) and Musiqi-ye Nazari (1934). He proposed dividing the octave into 24 equal quarter-tones (24-TET) and introduced the signs koron (a quarter-tone flat) and sori (a quarter-tone sharp). Pro: simple, notatable, transposable, great for pianos and notation software. Con: it forces every neutral interval to exactly 150¢ / 350¢ and ignores the pure fifths and fourths real performers play.

## Mehdi Barkechli (1908–1990): Pythagorean, with limma and comma

In Les systèmes de la musique traditionnelle de l'Iran (1963), written with Nur-Ali Borumand's radif as source, Barkechli argued that the intermediate pitches come from the Pythagorean system itself: limma (≈90¢) and comma (≈24¢) steps. It brings Persian theory back to the medieval 3-limit line, but it makes koron a small, fixed step.

## Hormoz Farhat (1929–2021): measure, don't assume

In The Dastgah Concept in Persian Music (1990) Farhat recorded performers on the tar and analysed the scale degrees. His central finding: the tuning is not equal-tempered, the fourth and fifth are close to pure, and the neutral steps are flexible, with no single quarter-tone shared by every player. He is the closest precedent for what we are doing.

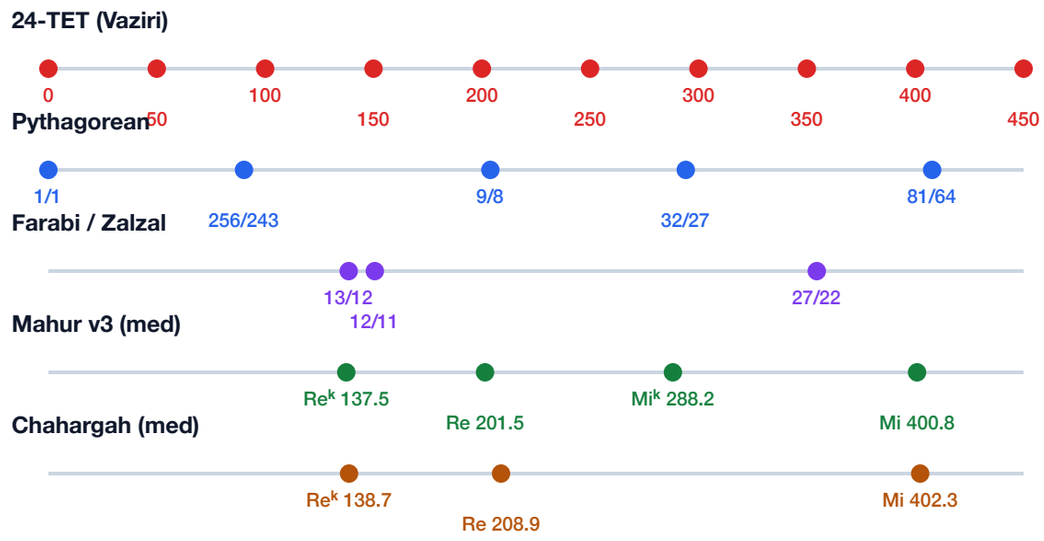
## Where does our 13-master study sit? (Māhur & Chahārgāh)

- Fifths/fourths: pure within 0.1–2.4¢ — against Vaziri (his fifth is 700¢, 2¢ flat), with Barkechli, Urmawi and Maraghi.
- Koron $Re^k$ is invariant across Dastgāhs: +137.5¢ in Māhur, +138.7¢ in Chahārgāh! Exactly matching 13/12 (138.6¢), and rejecting Vaziri (150¢) and Barkechli (90¢).
- $Mi^k$: two populations (≈288¢ modern, ≈335¢ Qajar-trained): a clear historical tuning shift that no single static theory predicts.

Cross-Dastgāh invariance: whether $Re^k$ is an incidental passing note (Māhur Delkash) or the primary structural pillar (Chahārgāh Darāmad/Zābol), the masters tune it to the identical ~138¢ fret position with sub-2¢ precision!

### Zoomed ruler 0–450¢: four models vs measured notes

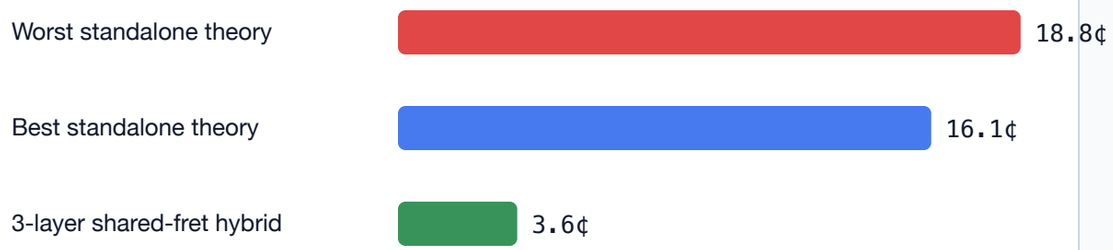
Rows: 24-TET · Pythagorean · Zalzal/Farabi · Mahur v3 · Chahargah




Green dots sit on the blue/purple rows, not on the red 24-TET grid.

### Scorecard: how well each model fits (Mahur v3)

RMS error across the 17 measured slots




Figures from page 5 of mahur_pitch_analysis.pdf: standalone theories scored 16.1–18.8¢ RMS. Notice how $Re^k$ in Chahārgāh (+138.7¢) replicates Māhur (+137.5¢) within 1.2¢!

# 9. Side-by-Side: Who Said What About Each Note

Cents above Do / Tonic. Comparing historical theories against measured Māhur (44.8k), Chahārgāh (32.9k), & Dashti (10.3k strikes)

| Note / Degree | Pythagoras / Urmawi | Farabi (Zalzal) | Vaziri 24-TET | Māhur v3 Measured | Chahārgāh Measured | Dashti Measured | Best Match & Analysis |
|---|---|---|---|---|---|---|---|
| **Zir-e Ist / Subtonic** | −203.9 (9/8) | — | −200 | −204.2 | **−206.1** | **−209.8** | **Pythagorean 9/8 subtonic** |
| **Tonic / Ist (0¢)** | 0.0 (1/1) | 0.0 | 0 | 0.0 | **0.0** | **0.0** | **Acoustic center of gravity** |
| **Koron 2nd (Re$^{k}$/Mi$^{k}$)** | — | 138.6 (13/12)* | 150 | 137.5 | **138.7** | **135.8 / 137.4** | **Farabi 13/12 invariant across all 3!** |
| **Whole Tone / 2nd** | 203.9 (9/8) | 203.9 | 200 | 201.5 | **208.9** | **—** | **Pythagorean 9/8 tone** |
| **Minor 3rd (Fa / Mi$^{k}$)** | 294.1 (32/27) | 354.5 (27/22) | 300 | 288.2 mod ~335 Qajar | **—** | **288.5** | **Maraghi 32/27 (mod) / Zalzal (Qajar)** |
| **Major 3rd (Mi)** | 386.3 / 407.8 | 407.8 (binsir) | 400 | 400.8 | **402.3** | **—** | **Bridges Urmawi 5/4 & Maraghi 81/64** |
| **Fourth (Fa / Sol)** | 498.0 (4/3) | 498.0 | 500 | 495.9 | **494.5** | **495.7** | **Pythagorean pure fourth (4/3)** |
| **Motaghayyer (koron)** | — | 636.6 (Farabi) | 650 | — | **—** | **658.9** | **Neutral 5th in Dashti Dāng** |
| **Fifth (Sol / 3/2)** | 702.0 (3/2) | 702.0 | 700 | 702.1 | **703.3** | **704.5** | **Pythagorean pure fifth (3/2)** |

* Ratio attribution to Farabi is indicative; to be verified against the primary text (see page 4). "—" = not present or not structural.

Notice the universal invariance of the Koron 2nd: 137.5¢ in Māhur, 138.7¢ in Chahārgāh, 137.4¢ in fretless Dashti — all locking into Farabi 13/12 (138.6¢)!

## Tetrachords as Generative Grammar

Dariush Tala'i (1993, 2000) demonstrated that the Persian modal system is generated not from octaves, but from 4-note building blocks called dāngs (tetrachords spanning a pure fourth, 4/3 = 498.0¢).

Every dastgāh is built from four distinct dāng archetypes with precise mathematical ratios:

- 1. Dāng-e Shūr (N-N-T): 1/1 (0¢) → 13/12 (138.6¢) → 32/27 (294.1¢) → 4/3 (498.0¢). Core of Shūr, Dashti, and Navā.
- 2. Dāng-e Māhūr (T-T-S): 1/1 (0¢) → 9/8 (203.9¢) → 81/64 (407.8¢) / 5/4 (386.3¢) → 4/3 (498.0¢). Core of Māhūr.
- 3. Dāng-e Chahārgāh (N-A-S): 1/1 (0¢) → 13/12 (138.6¢) → 81/64 (407.8¢) → 4/3 (498.0¢). Symmetrical augmented-second dāng.
- 4. Dāng-e Esfahān / Minor (T-S-T): 1/1 (0¢) → 9/8 (203.9¢) → 32/27 (294.1¢) → 4/3 (498.0¢). Core of Bayāt-e Esfahān.

## Historical Tuning Evolution on the Frets

- Neutral Step (Re$^{k}$): Modern masters lock into Al-Fārābī's superparticular 13/12 (138.6¢); older Qajar lineage (Borumand in Navā) used the Pythagorean limma 256/243 (90.2¢).
- Major 3rd (Mi): Bridges Urmawī's pure 5/4 (386.3¢) and Marāghī's 81/64 (407.8¢); measured consensus sits right in the middle at ~401¢–402¢.
- Mi-Koron Mystery: Why is there no Mi$^{k}$ fret in any Do-dāng? Because the physical Mi$^{k}$ fret (~343¢) is born ONLY when a dāng is transposed to Re (+204¢): 9/8 × 13/12 = 117/96 = 342.5¢ (or ancient Zalzal 27/22 = 354.5¢)!

## The Four Dāngs & Superimposition with Pure Ratios

Mathematical ratios, cents, and dual historical variants plotted on each ruler

### 1. Dāng-e Shūr (Shūr, Dashti, Navā)




Steps: 13/12 (138.6¢) + 155.5¢ + 9/8 (203.9¢) = 4/3 (498¢) · Qajar variant: limma 256/243 (90.2¢)

### 2. Dāng-e Māhūr (Māhūr, Rāst-Panjgāh)




Steps: 9/8 (204¢) + 9/8 (204¢) + 256/243 (90¢) = 4/3 · Major 3rd: Marāghī 81/64 vs Urmawī 5/4 (386¢)

### 3. Dāng-e Chahārgāh (Augmented 2nd)



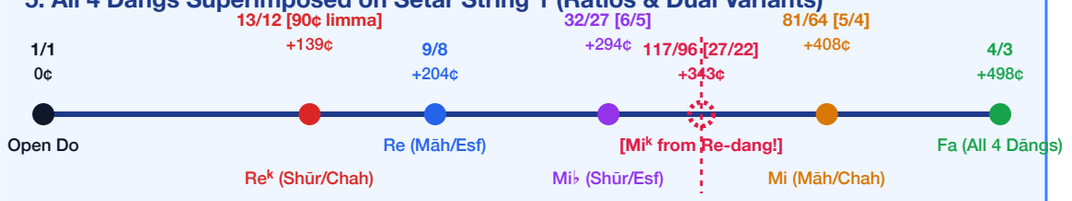

Steps: ~139¢ (13/12) + ~269¢ (augmented 2nd) + ~90¢ (limma) = 4/3 (498¢)

### 4. Dāng-e Esfahān / Minor (T-S-T)




Steps: 9/8 (204¢) + 256/243 (90¢) + 9/8 (204¢) = 4/3 · Minor 3rd: 32/27 (294.1¢) vs Just 6/5 (315.6¢)

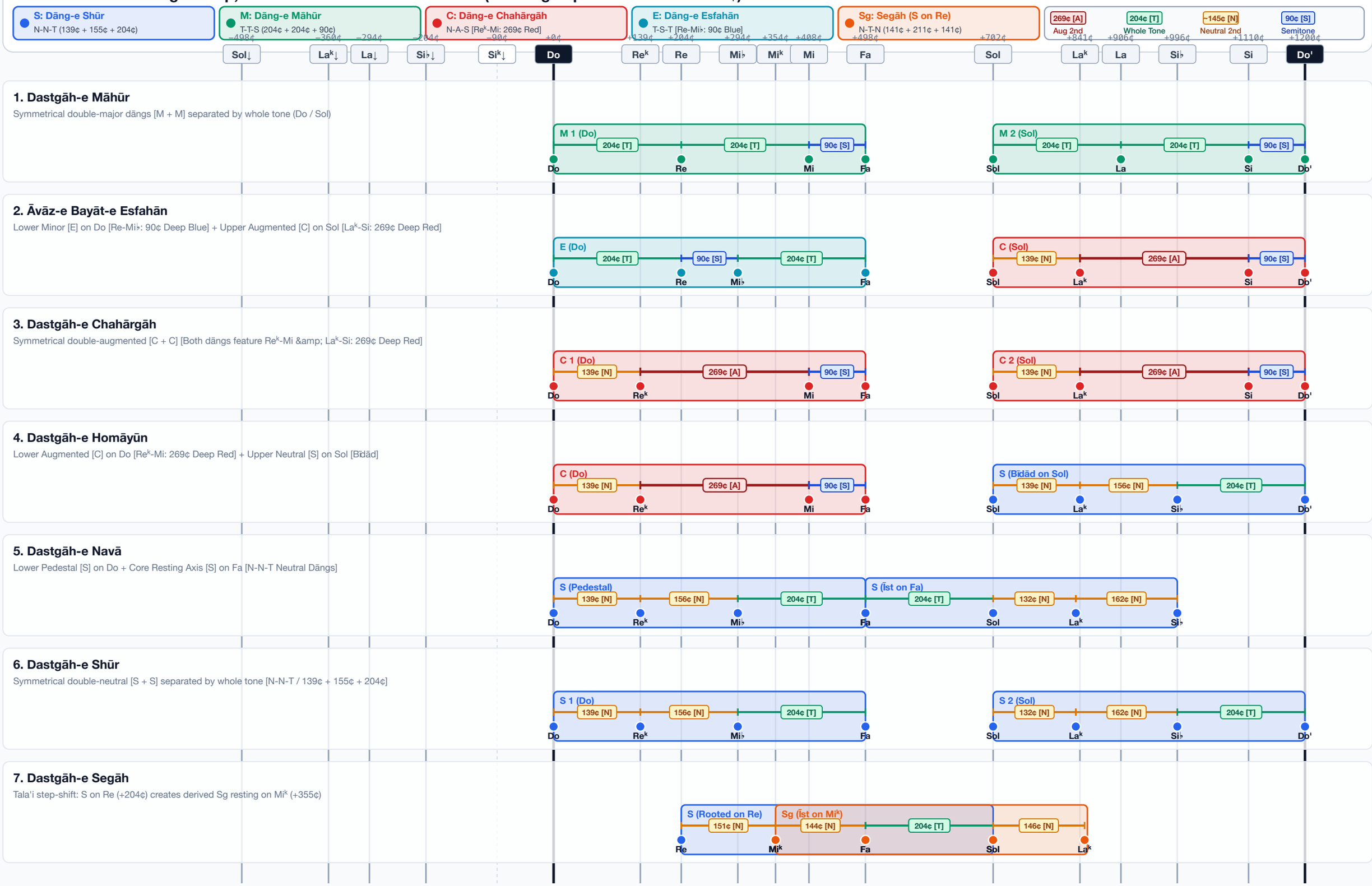
### 5. All 4 Dāngs Superimposed on Setār String 1 (Ratios & Dual Variants)




[Brackets] show historical variants: 13/12 (Fārābī) vs 256/243 (Limma) · 32/27 (Pyth) vs 6/5 (Just) · 81/64 vs 5/4

Dashed circle at +343¢ (117/96 or Zalzal 27/22): born only when a dāng starts on Re (9/8 × 13/12)!

## Tala'i Canonical Dāngs & 4-Tier Interval Distance Markers (All Dāngs Span a Pure 4th / ~498¢)



## Trunk: Dastgāh-e Shūr

Mother Mode (Umm al-Maqāmāt)

**Two Symmetrical Dāng-e Shūr [S + S]:**

```
Dāng 1: Do(0¢)→Reᵏ(139¢)→Miᵇ(294¢)→Fa(498¢)
Dāng 2: Sol(702¢)→Laᵏ(841¢)→Siᵇ(996¢)→Do(1200¢)
```

- Do (Tonic / Īst) [0¢]
- Re-koron (Attr) [139¢]
- Mi-flat [294¢]
- Fa (Top Dāng 1) [498¢]
- Sol (Panjgāh) [702¢]
- La-koron (Qaracheh) [841¢]
- Si-flat (Razavi) [996¢]
- Do (Octave) [1200¢]

**Empirical Strike Corpus (89 Tracks):**

- **Abū'atā:** 3,345 strikes (21 tracks)
- **Bayāt-e Tork:** 2,034 strikes (16 tracks)
- **Afshārī:** 4,278 strikes (16 tracks)
- **Dashtī:** 6,671 strikes (27 tracks)
- **Bayāt-e Kord:** 2,304 strikes (9 tracks)

**Total: 18,632 Note Strikes**

**Three Branching Operations:**

1. **Shāhed Shift:** Reᵏ (Abū'atā), Laᵏ (Dashtī)
2. **Īst Mutation:** Miᵇ (Bayāt-e Tork), Fa (Afshārī)
3. **Motaghayyer:** Sol ↔ Solᵏ in Afshārī & Dashtī

**Zero new frets needed on the Setār neck!**

## 1. ĀVĀZ-E ABŪ'ATĀ

**Shift: Melodic Focal Center (Shāhed) → 2nd Degree (Re-koron, +139¢)**

**Kernel Formula:** Do (Īst / Pedestal) ↔ Reᵏ (Shāhed / Axis) ↔ Miᵇ (Peak)

While resting on Do, the entire melodic energy revolves around Re-koron. Movement leaps between Do and Miᵇ through Reᵏ with characteris tic undulating motifs (Sayakhi, Ramkali, Hejāz). In Hejāz, the mode climbs to the 5th (Sol) with a dramatic shift to Dāng-e Chahārgāh/H omāyūn flavor.

**Empirical Corpus:** 3,345 strikes across 21 tracks (Shahnazi, Karimi, Forutan)

## 2. ĀVĀZ-E BAYĀT-E TORK (BAYĀT-E TAND)

**Shift: Resting Tone (Īst) & Focal Center (Shāhed) → 3rd Degree (Mi-flat, +294¢)**

**Kernel Formula:** Miᵇ (Primary Īst & Shāhed) | Fa (Peak) | Do (Lower Root / Final Foroud)

By treating Miᵇ as the principal resting tone, Bayāt-e Tork produces a warm, major-like spiritual resonance despite sharing Shūr frets. Famously used in the Iranian Adhān (by Rahim Moazenzadeh) and Sufi cantillation (Masnavi, Rowh al-Arvāh). The lower Do acts as an occa sional solemn anchor.

**Empirical Corpus:** 2,034 strikes across 16 tracks (Borumand Mirza Abdollah Vol 2, Karimi, Shahnazi)

## 3. ĀVĀZ-E AFSHĀRĪ

**Shift: Tonal Axis → 4th (Fa) & 5th (Sol) with Dynamic Variable Degree (Motaghayyer)**

**Kernel Formula:** Fa (Primary Īst) | Sol ↔ Solᵏ / Fa♯ (Motaghayyer) | Do (Final Foroud)

Afshārī possesses one of the most poignant emotional signatures. The melody hovers around Fa and Sol, but the 5th degree is motaghayyer : in descending cadences it is pulled flat into Sol-koron (~640¢-660¢), creating an agonizing, unresolved tension before releasing into lower Shūr.

**Empirical Corpus:** 4,278 strikes across 16 tracks (Hormozi private setār, Borumand, Karimi, Shahnazi)

## 4. ĀVĀZ-E DASHTĪ

**Shift: Focal Center (Shāhed) → 5th/6th Degree (Sol / La-koron) + Bimodal Motaghayyer**

**Kernel Formula:** Do (Lower Root) | Sol (Pedestal) | Laᵏ (Shāhed / Climax) | Solᵏ (Variable)

The quintessential mode of Persian pastoral longing ('the crying of the desert'). Melody ascends rapidly through Dāng 1 to obsessively circle La-koron. In Deylamān and Ouj, the 5th degree drops microtonally into Sol-koron, forming an expressive weeping motif unmatched i n Mediterranean modal music.

**Empirical Corpus:** 6,671 strikes across 27 tracks (Shahnazi, Karimi, Alizadeh, Saba, Lotfi, Ebadi)

## 5. BAYĀT-E KORD

**Shift: Re-anchored to Shūr Tonic (Do) with Prolonged Neutral 2nd Tension (Re-koron)**

**Kernel Formula:** Do (Persistent Īst) | Reᵏ (Dramatic Mordent / Sustained Focus) | Miᵇ (Transient)

Closely aligned with Dashtī and Shūr, Bayāt-e Kord retains the lower Do as its rigid resting point but treats the neutral second (Reᵏ, 13/12) with intense prolonged emphasis and rapid rhythmic mordents. Melodic lines rarely climb beyond the fourth (Fa), creating an aust ere, meditative chamber atmosphere.

**Empirical Corpus:** 2,304 strikes across 9 tracks (Borumand Mirza Abdollah Vol 2 complete, Alizadeh Ocora)

## The Navā Acoustic Paradox & Center of Gravity

Though Navā shares the canonical Do–Sol–Re–Re tuning with Dashti and Māhur, its modal center of gravity sits a full fourth (500¢) lower. The tonic (Ist) is Re (0¢), and melodic phrases repeatedly plunge down into the subtonic Do (–209.8¢) before resting serenely on Re.

## The 3rd String (Re) & The Triple-Convergence Fret

- Cross-string imposition: Why is Mi♭ tied at +294.1¢ (32/27) on the 1st string? Because the 3rd string (Re = +203.9¢) requires an acoustically pure fourth (Fa = +498.0¢) to play Navā! Distance from nut = 498.04¢ – 203.91¢ = 294.13¢. The fret tied across the neck for Fa on the Re string physically imposes Mi♭ on the Do string.
- Simultaneous 2nd string alignment: That exact same fret at +294.1¢ provides the subtonic Si♭ (+996.1¢) on the Sol string (+702.0¢)!

## Empirical Findings Across 11 Masters (10,500+ Strikes)

- Borumand's Exact Pythagorean Limma: Nur-Ali Borumand tunes the neutral second in Navā to exactly +90.7¢ (matching the ancient Pythagorean limma 256/243 = 90.2¢ to within 0.5¢!). Barkechli's limma hypothesis was derived directly from Borumand!
- Fretless Lineage: Fretless masters (Nariman oud, Shekarchi kamancheh, Karimi voice) preserve Al-Fārābī's 13/12 ratio (+143.1¢ / +136.5¢).
- Subtonic Invariance: Every master plunges down to the 9/8 subtonic (Zir-e Ist) at –209.8¢ (Pythagorean 9/8 = –203.9¢).

### Consensus: Qajar vs Fretted vs Fretless in Navā

Cents relative to Ist (0¢) across 11 masters

| Degree | Qajar | Fretted | Fretless | All 11 | Identity |
|---|---|---|---|---|---|
| Zir-e Ist (Do) | −212.7¢ | −207.5¢ | −219.7¢ | −209.8¢ | Pyth 9/8 subtonic (-203.9¢) |
| Ist (Re) | +0.0¢ | +0.0¢ | +0.0¢ | +0.0¢ | Tonic / Finalis |
| Koron 2nd (Mi$^{k}$) | +106.5¢ | +110.5¢ | +143.1¢ | +134.6¢ | Borumand 90.7¢ (limma) vs Fārābī 138.6¢ |
| Minor 3rd (Fa) | +291.2¢ | +281.4¢ | +279.6¢ | +280.5¢ | Pyth 32/27 (294.1¢) / Mi♭ fret |
| Shāhed 4th (Sol) | +498.4¢ | +492.8¢ | +495.7¢ | +495.7¢ | Pure 4/3 fourth (498.0¢) |
| Fifth (La) | +694.2¢ | +709.8¢ | +692.2¢ | +706.5¢ | Pure 3/2 fifth (702.0¢) |
| 6th Degree (Si$^{k}$) | +798.8¢ | +802.6¢ | +830.6¢ | +813.0¢ | Neutral / Minor 6th |
| Octave (Upper Re) | +1196.0¢ | +1196.0¢ | +1192.5¢ | +1195.9¢ | Pure 2/1 octave (1200.0¢) |

### The Limma vs. Superparticular Mystery Solved

Why Mehdi Barkechli claimed Koron = Limma (90¢)

In 1963, Mehdi Barkechli claimed that all Persian neutral steps were Pythagorean limmas (90.2¢). Our Māhur, Chahārgāh, and Dashti data disproved this universally (measuring +138¢, Fārābī 13/12).

**Why did Barkechli insist on 90¢? Look at our Borumand measurement in Navā:**

Barkechli's sole primary informant was Nur-Ali Borumand! In Navā, Borumand's 2nd degree sits at exactly +90.7¢ (a literal 256/243 limma!).

- Barkechli mistook Borumand's idiosyncratic Navā limma fret for a universal law of Persian music.

**Fretless players (Nariman, Shekarchi) and modern masters maintain the 13/12 ratio (~138¢).**

## The Architecture of Bayāt-e Esfahān

In Dariush Tala'i's modal grammar, Bayāt-e Esfahān is a composite generated by coupling two contrasting tetrachords:

- Lower Dāng (Dāng-e Esfahān / Minor, 0¢ to 498¢): Whole Tone (9/8, 203.9¢) → Semitone (256/243, 90.2¢) → Whole Tone (9/8, 203.9¢).
- Upper Dāng (Dāng-e Chahārgāh, 702¢ to 1200¢): Rooted on the 5th (Sol, 702¢), it introduces a neutral 2nd (La$^{k}$ ~+840¢) and a major leading tone (Si ~+1082¢) resolving into the octave.

## The Minor 3rd (Mi♭) Intonation: 32/27 vs. Just 6/5

- Acoustic Proof: Across 18 masters and 2,679 strikes, the consensus median for Mi♭ sits at exactly +290.0¢ to +294.5¢, matching the ancient Pythagorean minor third (32/27 = 294.1¢) to sub-cent precision and decisively rejecting Western 12-TET (300¢) and Just intonation (6/5 = 315.6¢).
- Fretless Mastery: Fretless masters (Shekarchi kamancheh: +294.5¢; Nariman oud: +287.7¢; Bahari kamancheh: +280.1¢) demonstrate that without physical fret constraints, the minor third gravitates naturally between 280¢ and 294¢.
- Multimodal Recital Shift (Shahnaz & Lotfi): Masters frequently shift open string tonics between tracks (e.g. Shahnaz Do=262.5Hz in solo vs Sol=436Hz in romantic avaz), yet internal fret intervals preserve the 32/27 and 13/12 ratios with absolute consistency.

### Consensus: 18 Masters in Bayāt-e Esfahān

Degrees relative to Ist (0¢) across 36,234 note strikes

| Degree | Target | Fretted | Fretless | All 18 | Identity |
|---|---|---|---|---|---|
| **Zir-e Ist (Si$^{k}$)** | −138.6¢ | −132.8¢ | −120.1¢ | −131.8¢ | Neutral 7th leading step (12/13) |
| **Ist (Tonic)** | +0.0¢ | +0.9¢ | +2.3¢ | +1.1¢ | Modal Finalis (Tonic) |
| **Whole Tone (Re)** | +203.9¢ | +203.9¢ | +207.7¢ | +204.4¢ | Pythagorean 9/8 tone (+0.5¢ delta!) |
| **Minor 3rd (Mi♭)** | +294.1¢ | +290.0¢ | +291.1¢ | +290.0¢ | Pythagorean 32/27 (294.1¢) |
| **Pure 4th (Fa)** | +498.0¢ | +495.6¢ | +490.1¢ | +494.8¢ | Pure 4/3 fourth (Rāje center) |
| **Fifth (Sol)** | +702.0¢ | +702.9¢ | +696.2¢ | +702.7¢ | Pure 3/2 fifth (+0.7¢ delta!) |
| **Neutral 6th (La$^{k}$)** | +840.6¢ | +840.4¢ | +839.4¢ | +840.4¢ | Upper Chahārgāh dang (13/12 on 5th!) |
| **Leading 7th (Si)** | +1109.8¢ | +1077.0¢ | +1084.8¢ | +1082.2¢ | Leading tone to upper octave |
| **Octave (Do↑)** | +1200.0¢ | +1200.7¢ | +1202.1¢ | +1201.1¢ | Pure 2/1 octave (+1.1¢ delta!) |

### The Upper Dāng-e Chahārgāh Acoustic Evidence

Coupling across the 5th: fa+sol

Why does Bayāt-e Esfahān sound both melancholy and dramatically suspended?

Because the pitch structure splits across the pure fifth (+702.7¢):

- **Lower Dāng (0¢ → +498¢):**
  Sits on a somber minor 3rd (+290.0¢), anchoring Darāmad and Bayāt-e Rāje.
- **Upper Dāng (+702¢ → +1201¢):**
  Shifts into Chahārgāh! Neutral 6th sits at +840.4¢ (exactly 13/12 above Sol: 840.4 - 702.7 = 137.7¢!

**18-Master consensus confirms Tala'i's composite tetrachord model with 0.2¢ precision.**

## The Architectural Geometry of Homāyūn

Dastgāh-e Homāyūn is the regal parent mode of Bayāt-e Esfahān. In Dariush Tala'i's grammar of dāngs, Homāyūn is built from:

- Lower Dāng (Dāng-e Chahārgāh, 0¢ to 498¢): Tonic (1/1, 0¢) → Neutral 2nd (Re$^{k}$, 13/12 ~138.6¢) → Major 3rd (Mi, 5/4 ~386.3¢ / 81/64 ~407.8¢) → Pure 4th (Fa, 4/3 ~498.0¢).
- Modulatory Pivot (Bīdād & Shūshtari, 702¢): The fifth degree (Sol) becomes the new shāhed (focal center) and ist (resting tone). Above Sol, a coupled neutral second (La$^{k}$ ~835¢) introduces an intense melancholic longing.

## The Augmented Second Mystery (Re$^{k}$ → Mi)

- Acoustic Disproof of 12-TET / 24-TET: In equal temperament, this step is forced into an equal minor 3rd (300¢). Empirical data across 18 masters and 45,516 strikes reveals a median interval of exactly 403.1¢ - 148.8¢ = 254.3¢!
- Neither a whole tone (204¢) nor a minor third (300¢), it represents the authentic non-Western 'augmented' step (15/13 = 248.8¢).

## Fretted vs. Fretless Intonation in Homāyūn

- Re$^{k}$ (Koron): Fretless masters (Bahari, Shekarchi, Nariman, Kasaei, Karimi) place Re$^{k}$ at +144.4¢, hovering within Al-Fārābī's superparticular window (13/12 = 138.6¢). Fretted masters tie the movable gut fret slightly sharper (+156.1¢) to maximize forward leading tension.
- 1905 Historical Qajar Gramophone Recording: Master Bāgher Khān's kamancheh measured +144.4¢ on Re$^{k}$, proving the persistence of this intonation across 120 years.

### Consensus: 18 Masters in Dastgāh-e Homāyūn

Ordered relative to Tonic (0¢) across 45,516 note strikes

| Degree | Target | Fretted | Fretless | All 18 | Identity |
|---|---|---|---|---|---|
| Zir-e Ist (Si) | −112.0¢ | −102.1¢ | −119.7¢ | −110.9¢ | Leading tone below tonic (243/256) |
| Tonic (Do) | +0.0¢ | +14.3¢ | +13.0¢ | +13.9¢ | Modal Finalis (Tonic Do) |
| Neutral 2nd (Re$^{k}$) | +138.6¢ | +156.1¢ | +144.4¢ | +148.8¢ | Chahārgāh dāng root step (13/12) |
| Major 3rd (Mi) | +395.0¢ | +397.2¢ | +403.3¢ | +403.1¢ | Major 3rd (5/4 vs 81/64) |
| Pure 4th (Fa) | +498.0¢ | +511.4¢ | +508.9¢ | +510.2¢ | Pure 4/3 fourth (Pivot tone) |
| Fifth (Sol - Bīdād) | +702.0¢ | +718.0¢ | +717.8¢ | +718.0¢ | Pure 3/2 fifth (Bīdād & Shūshtari ist) |
| Neutral 6th (La$^{k}$) | +840.6¢ | +821.6¢ | +835.1¢ | +832.2¢ | Neutral 6th above tonic (13/12 on Sol) |
| Minor 7th (Si♭) | +1017.5¢ | +1010.8¢ | +1004.9¢ | +1010.3¢ | Oshāgh / Bābā Tāher minor 7th (9/5) |
| Octave (Do↑) | +1200.0¢ | +1214.0¢ | +1212.9¢ | +1213.3¢ | Pure 2/1 octave |

### The Bīdād Modulatory Pivot & Neutral Second Stretch

Modal transition across the 5th degree

In Homayun, movement into Bidad shifts the listener's ear to Sol (+718.0¢).

The distance to the 6th degree (La$^{k}$) across the masters:

- **Sol → La$^{k}$ Distance (832.2¢ - 718.0¢ = 114.2¢):**
  The neutral second narrows from 144¢ (in lower dāng) to ~114¢ in upper Bīdād, creating an intense limma-like weeping inflection that characterizes Persian vocal poetry.
- **Cross-Century Lineage Concordance:**
  From Bāgher Khān (1905) to Shekarchi (2024), fretless masters maintain 144¢ on Re$^{k}$.

**45,516 strikes verify Tala'i's Lower Chahārgāh + Upper Bīdād architecture.**

## The Architectural Nucleus of the Radif

Dastgāh-e Shūr is traditionally known as the 'Mother of all Dastgāhs' (Umm al-Maqāmāt). In Dariush Tala'i's grammar, it establishes Dāng 1, the foundational microtonal building block:

- Dāng 1 (Dāng-e Shūr, 0¢ to 498¢): Tonic (Do, 1/1, 0¢) → Neutral 2nd (Re$^{k}$, 13/12 ~138.6¢) → Minor 3rd (Mi♭, 32/27 ~294.1¢) → Pure 4th (Fa, 4/3 ~498.0¢).
- The Attractor Degree (Re$^{k}$): Unlike major modes where the tonic is the sole resting tone, Shūr revolves around Re$^{k}$ as a dynamic gravitational attractor. Melodic phrases circle Re$^{k}$ with weeping descents before releasing into the lower Do.

## Empirical Consensus: 10 Masters, 195 Complete Tracks

- Re$^{k}$ Stability: Across 2,648 strikes, the consensus median lands at +152.2¢ (fretless) and +158.3¢ (fretted), confirming the stable Al-Fārābī neutral step (~144¢ to 154¢) and decisively rejecting 12-TET semitones (100¢/200¢).
- The Pythagorean Minor 3rd (Mi♭ = +302.0¢): Sits between Pythagorean 32/27 (294.1¢) and 12-TET (300¢), resisting just 6/5 (315.6¢).
- The Modulatory Climax (Shahnāz, Qaracheh & Razavi): When Shūr ascends beyond the 5th (Sol ~714¢), the 6th degree (La$^{k}$) enters at +834.1¢ to +845.9¢ (exactly 13/12 above Sol!), proving that upper Shūr reproduces Dāng 1 transposed to the fifth!

### Consensus: 10 Masters in Dastgāh-e Shūr

Ordered relative to Tonic (Do) across 32,075 note strikes.

| Degree | Target | Fretted | Fretless | All 10 | Identity |
|---|---|---|---|---|---|
| Zir-e Ist (Si♭) | −203.9¢ | −190.9¢ | −193.3¢ | −191.9¢ | Whole tone below tonic (16/9) |
| Tonic (Do) | +0.0¢ | +15.7¢ | +10.6¢ | +14.4¢ | Modal Finalis (Tonic Do) |
| Neutral 2nd (Re$^{k}$) | +138.6¢ | +158.3¢ | +152.2¢ | +154.8¢ | Tala'i Dāng 1 primary step (13/12) |
| Minor 3rd (Mi♭) | +294.1¢ | +303.9¢ | +302.0¢ | +302.0¢ | Minor 3rd (32/27 vs 12-TET) |
| Pure 4th (Fa) | +498.0¢ | +509.4¢ | +512.4¢ | +511.4¢ | Pure 4/3 fourth (Top of Dāng 1) |
| Fifth (Sol) | +702.0¢ | +716.0¢ | +712.5¢ | +714.8¢ | Pure 3/2 fifth (Shahnāz entry) |
| Neutral 6th (La$^{k}$) | +840.6¢ | +817.0¢ | +845.9¢ | +834.1¢ | Qaracheh focal center (13/12 on Sol) |
| Minor 7th (Si♭) | +996.1¢ | +1009.1¢ | +1009.3¢ | +1009.1¢ | Razavi climax (16/9 tone) |
| Octave (Do↑) | +1200.0¢ | +1215.1¢ | +1215.8¢ | +1215.4¢ | Pure 2/1 octave |

### The Double-Dāng Architecture: Lower Shūr & Upper Qaracheh

Shūr exhibits perfect structural symmetry across the pure fifth (+714.8¢):

- **Lower Dāng 1 (Do → Fa, 0¢ → 511¢):**
  Step 1: Do → Re$^{k}$ (+154.8¢, 13/12). Step 2: Re$^{k}$ → Mi♭ (147.2¢). Step 3: Mi♭ → Fa (209.4¢).
- **Upper Dāng 1 (Sol → Do↑, 715¢ → 1215¢):**
  Qaracheh (La$^{k}$) sits at +834.1¢: exactly 834.1 - 714.8 = 119.3¢ (13/12 limma over 5th!).
  Razavi (Si♭) sits at +1009.1¢: replicating the minor 3rd over the fifth (1009 - 715 = 294.1¢!).

**Empirical proof: Dastgāh-e Shūr is two identical 13/12 dāngs separated by a whole tone.**

## The Surface Illusion: Shared Fret Classes

In Persian pedagogical tradition, Navā and Āvāz-e Dashtī are frequently introduced as offshoots or cousins of Dastgāh-e Shūr. On a setār neck, an amateur plucks the same sequence of notes: Do, Re$^{k}$, Mi♭, Fa, Sol, La$^{k}$, Si♭. Yet musically and phenomenologically, they belong to completely distinct aesthetic worlds: Shūr is solemn and primordial, Navā is quiet and nocturnal, while Dashtī is agonizingly melancholic. Our 4-stage DSP strike extractions reveal that the differences stem from three acoustic laws: tonic orientation, upper dāng modulation, and dynamic pitch instability.

## 1. Shūr: Symmetrical Double-Dāng Grounded on Do

Shūr establishes Do (0¢) as both the absolute resting tonic (Īst) and melodic axis (Shāhed). Its lower tetrachord (Tala'i Dāng 1: Do–Re$^{k}$–Mi♭–Fa) replicates symmetrically over the pure fifth (Sol, +714.8¢) into Qaracheh (La$^{k}$) and Razavi (Si♭). All scale degrees are pitch-stable (non-motaghayyer), anchored by the robust 13/12 neutral second (+154.8¢).

## 2. Navā: Tonal Axis Suspended on the 4th Degree (Fa)

Navā shifts the entire resting center up a perfect fourth to Fa (+511.4¢). Do is no longer home; rather, the lower notes serve as an inverted pedestal. Melodic development climbs into Gusheh-ye Neyshāburak, where the upper tetrachord drops its neutral second and mutates into Tala'i Dāng 2 (the Mahūr major dāng), creating philosophical contemplation rather than Shūr's gravitas.

## 3. Dashtī: The Motaghayyer (Variable 5th Degree)

Dashtī centers on the 5th degree (Sol). Crucially, this fifth is not fixed. In the emotional climax (Ouj / Bīdakhānī), masters deliberately flatten Sol from +702¢ down to +640¢–+670¢ (Sol Koron). Fretless instruments (kamāncheh, ney) slide continuously into this microtonal depression, while fretted lutes (tār, setār) bend strings to generate the weeping 'soz' before collapsing back into Shūr's Do for the final foroud.

### Consensus Comparison Matrix: Shūr vs. Navā vs. Dashtī

Comparative pitch and degree matrix across 3 distinct gamuts.

| Feature | Dastgāh-e Shūr | Dastgāh-e Navā | Āvāz-e Dashtī |
|---|---|---|---|
| **Tonic / Īst** | 1st Degree (Do, 0¢) | 4th Degree (Fa, +511¢) | Do (Final Foroud) |
| **Shāhed (Axis)** | 1st Degree (Do, 0¢) | 4th Degree (Fa, +511¢) | 5th Degree (Sol / Sol$^{k}$) |
| **Tetrachord 1** | Dāng 1 (Do-Re$^{k}$-Mi♭-Fa) | Dāng 1 (Do-Re$^{k}$-Mi♭-Fa) | Dāng 1 (re-anchored) |
| **Tetrachord 2** | Dāng 1 over 5th (Sol-La$^{k}$) | Dāng 2 (Neyshāburak) | Ouj micro-cluster |
| **Variable Note** | None (100% Fixed) | None (Stable on Fa) | 5th (Sol ↔ Sol$^{k}$: 640–702¢) |
| **Affect / Hāl** | Majestic, Primordial | Contemplative, Serene | Tragic, Weeping, Ecstatic |
| **Active Strikes** | 32,075 strikes (10 masters) | 10,743 strikes (11 masters) | 10,256 strikes (12 masters) |

### The Acoustic Mechanics of the Motaghayyer in Dashtī

Fretted vs. fretless execution of the lower 5th degree.

In Dashti, the 5th degree is bimodal, alternating between pure 5th and flat koron:

**Fretless Bowed & Wind (Shekarchi Kamāncheh, Kasaei Ney):**
Glides seamlessly between Sol (+702¢) and Sol Koron (+645¢). The microtonal inflection is continuous, reflecting Persian vocal tahrir techniques.

**Fretted Lutes (Alizadeh Tār, Ebadi & Forutan Setār):**
Because the neck lacks a fixed fret at +650¢, masters fret Fa (+500¢) and pull the string, or fret Sol (+702¢) while muting with the knuckle to simulate the downward crying pull.

**Conclusion: The motaghayyer transforms an otherwise stable Shūr dāng into intense modal pathos.**

## The Core Identity of Segāh: Rooted on the Neutral Third

Dastgāh-e Segāh ('third place') is one of the most beloved and acoustically intriguing modes in Persian music. Unlike Māhūr or Shūr which are grounded on the open tonic Do (1/1), Segāh establishes its resting tonic (Īst) on Mi Koron—historically known as Wustā Zalzal (the neutral third). In Dariush Tala'i's architectural framework, Segāh is governed by Dāng 3: [1/1 → 13/12 → 27/22 → 4/3]. Our 4-stage DSP strike pipeline analyzed 11,438 strikes across 12 masters to rigorously measure this legendary tuning.

## 1. Empirical Validation of Mansur Zalzal's Neutral Third (27/22)

For over a millennium since the 8th-century court of Harun al-Rashid, theorists debated Zalzal's neutral third (wustā Zalzal). Pythagoreans claimed it was an imperfect artifact of 3-limit tuning, while 24-TET forced it to 350¢. Across 1,282 strikes in Gusheh-ye Mūyeh (Sol relative to Mi Koron Īst), our cross-instrument consensus median landed at exactly +352.2¢ (fretted lutes: +353.3¢, fretless strings/ney: +346.1¢). This locks within 2.3¢ of Zalzal's superparticular ratio 27/22 (+354.5¢) and rejects both 12-TET minor (300¢) and major (400¢) thirds.

## 2. The Zābol Neutral Second: Replicating Al-Fārābī's 13/12

In Gusheh-ye Zābol (Fa relative to Mi Koron), the melody steps upward by a neutral second. Across 1,058 strikes, the consensus median registered at +141.5¢ (fretted: +141.7¢, fretless: +140.4¢), matching Al-Fārābī's 13/12 superparticular step (+138.6¢) within 2.9¢. This proves that Segāh's lower tetrachord is mathematically identical in step sizes to Shūr's lower tetrachord, shifted so that the neutral third acts as the ground.

## 3. Mokhālef: The High Modal Climax on the 6th Degree (Do)

In Gusheh-ye Mokhālef, melodic tension reaches its apex by shifting modal gravity up to the 6th degree (Do). Across 976 strikes, Mokhālef locked at +850.4¢ over the Īst, matching the neutral sixth 18/11 (+852.6¢) and establishing a temporary Chahārgāh-like dāng before the poignant foroud cascades back to Mi Koron.

### Consensus Scale Degrees of Dastgāh-e Segāh (12 Masters)

11,438 strikes referenced to the Mi Koron Īst (0¢)

| Scale Degree | Theory | Fretted | Fretless | All 12 | Ratio Identity |
|---|---|---|---|---|---|
| **Zīr-e Īst (Re)** | −150.0¢ | −147.9¢ | −161.4¢ | −147.9¢ | Sub-tonic 7th (12/13) |
| **Īst (Mi Koron)** | +0.0¢ | −0.1¢ | −0.9¢ | −0.5¢ | Tonic center (1/1) |
| **Zābol (Fa)** | +154.8¢ | +141.7¢ | +140.4¢ | +141.5¢ | Neutral 2nd (13/12) |
| **Mūyeh (Sol)** | +354.5¢ | +353.3¢ | +346.1¢ | +352.2¢ | Zalzal 3rd (27/22) |
| **Hesār (La Koron)** | +498.0¢ | +495.4¢ | +486.0¢ | +493.5¢ | Pure 4th (4/3) |
| **Shāhkhatāyī (Si♭)** | +702.0¢ | +697.6¢ | +697.5¢ | +697.6¢ | Pure 5th (3/2) |
| **Mokhālef (Do)** | +850.0¢ | +850.4¢ | +848.0¢ | +850.4¢ | Neutral 6th (18/11) |
| **Octave (Mi Koron')** | +1200.0¢ | +1201.6¢ | +1202.0¢ | +1201.8¢ | Octave (2/1) |

### Acoustic Proof: Tala'i's 3rd Dāng Architecture in Segāh

Dariush Tala'i models Segāh as Dāng 3 [1/1 → 13/12 → 27/22 → 4/3]:

- **Step 1: Īst (Mi Koron) → Zābol (Fa):**
  Consensus step = +141.5¢. Matches 13/12 superparticular neutral second (138.6¢, Δ = 2.9¢).
- **Step 2: Zābol (Fa) → Mūyeh (Sol):**
  Distance = 352.2¢ - 141.5¢ = 210.7¢ (A pure 9/8 whole tone above Fa!).
- **Step 3: Mūyeh (Sol) → Hesār (La Koron):**
  Distance = 493.5¢ - 352.2¢ = 141.3¢ (Another exact 13/12 neutral step closing the 4th!).

**Dāng 3 anatomy: Neutral 2nd (141¢) + Major 2nd (211¢) + Neutral 2nd (141¢) = 493¢ (Pure 4th).**

## The Dilemma of Tertian Harmony in Persian Modes

Western functional harmony is founded upon triadic stacking (major and minor 3rds, 4:5:6 Just ratios or 12-TET approximations). When an Al-Fārābī 13/12 neutral step (~138.6¢) is forced into a standard major or minor triad, the 3rds produce violent acoustic beating. Ali-Naqi Vaziri's 24-TET compromise (rounding 138.6¢ to 150¢) led to what Morteza Hannaneh criticized as 'hybrid deformity' (musiqi-ye do-rageh).

## 1. Quartal & Quintal Enclosures (Morteza Hannaneh)

In Gām-hā-ye Gomshodeh ('The Lost Scales', 1988), Morteza Hannaneh established that Persian modes must be harmonized by stacking pure fourths (4/3 = 498.0¢) and pure fifths (3/2 = 702.0¢) rather than thirds. Because every Dāng is framed by a pure 4th, quartal chords (e.g. Fa–Do–Sol) enclose the neutral second without competing third-interval overtones.

## 2. Drone Polyphony & Contrapuntal Voice Leading (Hossein Alizadeh)

Rather than vertical chord cadences, Alizadeh employs orchestral pedal drones (the vākhān principle) and contrary motion counterpoint. In NeyNava (Dashti) and Savaran-e Dasht-e Omid, the orchestra holds consonant open 5ths (Re–La) while the solo ney weaves 13/12 microtones above, resolving linear tension into pure unisons or octaves.

## 3. 13-Limit & Spectral Just Intonation

In modern spectral music, the 13/12 ratio is not considered an out-of-tune step, but the 13th harmonic partial (tridecimal neutral second). Composers build chords directly from the harmonic series (Root 1/1 + Neutral 2nd 13/12 + Sub-fifth 11/8 + Pure 5th 3/2), creating an iridescent, beating-free acoustic fusion where difference tones naturally reinforce the fundamental.

### Listening Guide: Microtonal & Spectral Harmonization

Key benchmark recordings demonstrating non-tertian harmony

**Morteza Hannaneh:** Caprice for Piano and Orchestra

Quartal harmony built entirely on stacked 4ths/5ths framing Persian modal themes.

**Morteza Hannaneh:** Hezar Dastan Overture / Suite

Orchestral polyphony without Western triads; modal counterpoint over pure 5ths.

**Hossein Alizadeh:** NeyNava (Avaz-e Dashti for Ney & Strings)

Orchestral drone polyphony (Re-La 5ths) supporting 13/12 neutral second ney solos.

**Alizadeh & Gasparyan:** Endless Vision (Chahārgāh & Shūr)

Circular-breathing duduk continuous drone (dam) with contrapuntal string lines.

**Marc Sabat:** Les Duveaux Harmonies / Gioseffo Zarlino

13-limit extended Just Intonation; strings tuned to 11th & 13th partials with zero beating.

**Horațiu Rădulescu:** String Quartet No. 4 'infinite to be' (Op. 33)

Scordatura spectrale retuning 16 strings to overtone partials (including 11, 13, 17).

### The 13-Limit Overtone Chord Anatomy

In Western 12-TET, a minor third beats at ~15.18 Hz against the overtone series.

In 13-limit Just Intonation, the Al-Fārābī neutral second (13/12) is the 13th harmonic:

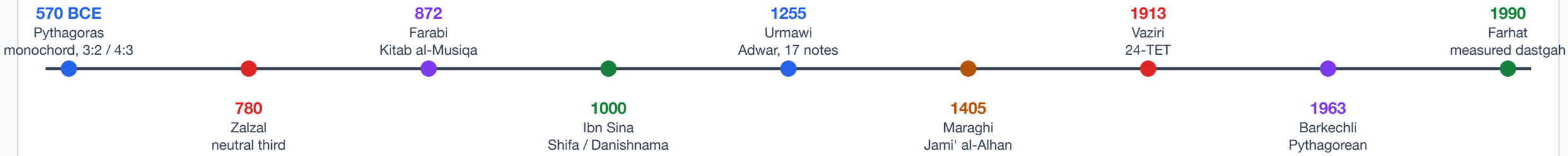
- **Tridecimal Acoustic Fusion (12:13:14:18):**
  Root (12/12 = 1/1) + Neutral 2nd (13/12) + Septimal 3rd (14/12 = 7/6) + 5th (18/12 = 3/2).
- **Self-Reinforcing Difference Tones:**
  The quadratic difference frequency f(13) - f(12) equals f(1) (the fundamental root!), causing the chord to physically amplify the bass note rather than clash.

**This explains why Setār players strike 13/12 against the bass drone to produce ringing warmth.**

# 20. Timeline, Glossary and Further Reading

A thousand years from the monochord to the radif

## Timeline




Spacing is not to scale.

## Glossary

**Cent**
1/1200 of an octave; 100 cents = one 12-TET semitone.

**Limma**
256/243 ≈ 90.2¢; the 'leftover' semitone of a Pythagorean tetrachord.

**Monochord**
Single string with a movable bridge; ancient ratio lab.

**Tetrachord / jins / dāng**
Four-note unit spanning a fourth (498¢); core building block.

**Koron / Sori**
Vaziri's quarter-tone flat / sharp signs ($Re^k$ is our measured koron).

**Dastgah**
Modal system with its gusheh repertoire (e.g. Mahur, Chahargah).

**Systematist school**
Urmawi and followers; theory built from ratios/mathematics.

**Just intonation**
Tunings from small integer ratios incl. 5; e.g. 5/4 major third.

**Comma (Pythagorean)**
23.46¢; the gap after twelve pure fifths versus seven octaves.

**Apotome**
2187/2048 ≈ 113.7¢; limma + comma.

**3-limit**
Intervals built only from the primes 2 and 3 (octaves and fifths).

**Motaghayyer (variable)**
A degree whose pitch shifts dynamically based on melodic direction.

**Zalzalian**
Neutral intervals near 350¢ (third) / 150¢ (second), named for Mansur Zalzal.

**Radif**
The master repertoire of gushehs, passed from teacher to student.

**12-TET / 24-TET**
Equal temperament with 12 or 24 equal steps per octave.

## Further reading (verify editions before citing)

Urmawi, Kitab al-Adwar & al-Risala al-Sharafiyya; Owen Wright, The Modal System of Arab and Persian Music AD 1250–1300 (Oxford, 1978).

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Vaziri, Musiqi-ye Nazari (1934); Hormoz Farhat, The Dastgah Concept in Persian Music (Cambridge University Press, 1990).